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Are your foods free from **AFLATOXIN ???**



AFLATOXIN DETECTION AND HEALTH RISK ASSESSMENT

Problem

We are eating a variety of foods in our daily life but those foods can be hazardous for our health if those foods are contaminated with substances like **Aflatoxins**

In many developing regions, including Uganda, routine laboratory testing for aflatoxin contamination remains inaccessible, slow, or unaffordable for smallholder farmers, grain traders, and local food safety inspectors, leaving large volumes of cereal produce unscreened before reaching markets and households.

▪ Aflatoxins

Aflatoxins are poisonous carcinogens that are produced by certain molds 'Aspergillus flavus' and 'Aspergillus parasiticus'. Aflatoxins are regularly found in improperly stored in cereals (maize, sorghum, rice) oil seeds (peanuts, cotton seed) spices and tree nuts, particularly in warm, humid regions

The mold can colonize and contaminate food before harvest or during storage, especially following prolonged exposure to a high humidity environment or stressful conditions such as drought.

Health risks

Chronic dietary exposure has been linked to liver damage, immune suppression, and impaired growth in children (World Health Organization, 2023), and aflatoxins have been classified as a Group 1 human carcinogen on the basis of their established association with liver cancer (International Agency for Research on Cancer, 2012).

Solution; AflAlert

What it does

AflAlert is a specialized mobile diagnostic application designed to evaluate aflatoxin contamination risks in **maize and groundnut** value chains. The application removes the need for hardware-heavy testing by combining edge-based computer vision with smartphone-assisted colorimetric strip reading.

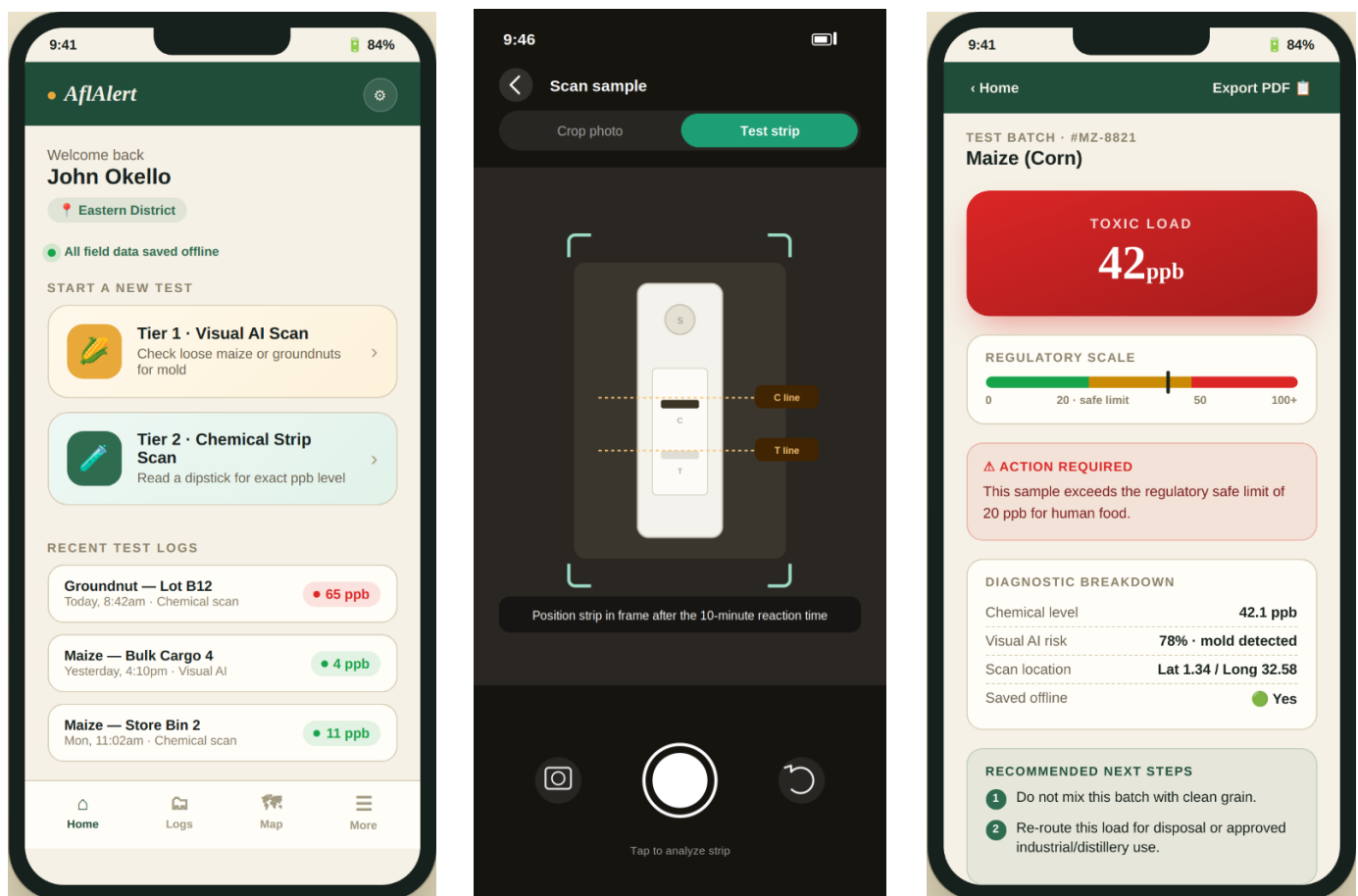
The app uses a specialized AI eyeball trained specifically on photos of corn and peanuts. When you show it a picture of these crops, it can instantly spot if they are moldy, damaged by bugs, or broken apart. It also functions as a digital reader for **commercially available lateral flow paper strips**. Following an on-site liquid extraction process, the user captures an image of the reacted test strip. The app's image-processing algorithms isolate the test and control lines, calculate their optical density ratio, and output a precise, quantitative contamination reading in parts per billion (ppb). This system transforms standard smartphones into mobile agricultural laboratories, providing smallholder farmers and traders with immediate food safety data

Relevance

- Reduce consumer exposure to toxic aflatoxin loads in staple foods such as maize and groundnuts.
- Farmers can identify contaminated grain at the farm gate, preventing mixing with clean produce and reducing whole-batch rejection at markets.
- Providing a low-cost, smartphone-based diagnostic that replaces expensive laboratory equipment costing a lot of money.

- Empowering farmers with data-driven decisions at the point of harvest, reducing dependency on expensive third-party testing services.
- By leveraging edge-based AI computer vision, the app delivers laboratory-grade analysis in field conditions, even with limited internet connectivity.
- The result is a scalable diagnostic tool that can rapidly penetrate rural agricultural communities across Uganda
- Enabling agri-exporters to conduct pre-shipment self-assessments, reducing the likelihood of costly border rejections.
- Contributes to Sustainable Development Goal 2 (Zero Hunger) and SDG 3 (Good Health and Well-Being) by directly addressing food safety at the grassroots level (UN, 2015).

Prototypes



References

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- World Bank. (2014). Missing food: The case of postharvest grain losses in Sub-Saharan Africa. Report No. 60371-AFR. Washington, D.C.: World Bank.
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